

NETWORK MONITORING & FIREWALL CONFIGURATION LAB

Practical Networking and Cybersecurity Project
Report

Tools Used

Cisco Packet Tracer • Wireshark • Kali Linux • UFW

Topics Covered

VLANs • Inter-VLAN Routing • OSPF • ACLs • Packet
Analysis • Linux Firewall

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1. Introduction

Network monitoring is an important part of enterprise network administration and cybersecurity. In this practical lab, a small routed network was designed and configured using Cisco Packet Tracer. The objective was to understand how routers and switches work together, how traffic is separated using VLANs, how routing information is exchanged through OSPF, and how access control can be applied using an extended ACL. The lab also included Wireshark analysis to observe real network traffic and Kali Linux firewall configuration using UFW.

The practical approach was useful because each configuration was verified through command-line output rather than relying only on configuration commands. This makes the work closer to a real network troubleshooting and monitoring workflow. Screenshots were collected as evidence for the configuration and testing stages.

2. Objectives

- Design and configure a small routed enterprise-style network.
- Create VLANs and implement logical network segmentation.
- Configure inter-VLAN routing using router subinterfaces.
- Configure and verify OSPF dynamic routing.
- Implement an extended ACL to restrict selected inter-network traffic.
- Perform connectivity and ACL verification tests.
- Capture and analyze ICMP, TCP, DNS and QUIC traffic in Wireshark.
- Configure and verify a host firewall using Kali Linux UFW.

3. Lab Environment

The practical environment consisted of Cisco Packet Tracer for the simulated network, Wireshark for packet capture and protocol analysis, and Kali Linux for host firewall and security-log related work. Screenshots in the following sections provide evidence of the configuration and verification steps performed during the lab.

4. Router Initial Configuration

The first stage of the lab involved basic router management configuration. A hostname was assigned to identify the device, and an enable secret was configured to protect privileged EXEC access. Basic management configuration is important because network devices should be identifiable and privileged access should not be left unprotected.

The router CLI provides direct control over interfaces, routing protocols, security policies and operational verification commands. During the configuration process, the command-line interface also displayed IOS prompts and system messages. These messages are useful during troubleshooting because they can indicate interface state changes, configuration mode, and protocol events.

In an enterprise environment, basic device hardening normally continues with secure management protocols, local user accounts, appropriate privilege levels, logging and restricted management access. In this lab, the focus was on establishing the basic configuration required for the remaining networking tasks.

```
--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]:
% Please answer 'yes' or 'no'.
Would you like to enter the initial configuration dialog? [yes/no]: yes

At any point you may enter a question mark '?' for help.
Use ctrl-c to abort configuration dialog at any prompt.
Default settings are in square brackets '[]'.

Basic management setup configures only enough connectivity
for management of the system, extended setup will ask you
to configure each interface on the system

Would you like to enter basic management setup? [yes/no]: yes
Configuring global parameters:

Enter host name [Router]: sach

The enable secret is a password used to protect access to
privileged EXEC and configuration modes. This password, after
entered, becomes encrypted in the configuration.
Enter enable secret: Sach

The enable password is used when you do not specify an
enable secret password, with some older software versions, and
some boot images.
Enter enable password:
```

Copy

Figure 1: Router Initial Configuration / Basic Management Setup

5. Switch Trunk Verification

VLANs provide logical separation of devices on a switched network. Instead of placing every endpoint into one broadcast domain, different groups can be assigned to different VLANs. In this lab, VLAN 10 was used for the ADMIN network and VLAN 20 for the USERS network.

The switch ports connected to end devices were configured as access ports and assigned to their respective VLANs. The link between the switch and router was configured as a trunk so that traffic belonging to multiple VLANs could travel over the same physical connection. IEEE 802.1Q tagging allows the receiving device to identify the VLAN associated with each frame.

VLAN segmentation improves organization and can also support security policies. For example, administrative systems can be separated from normal user systems and traffic between them can later be controlled with routing and ACL rules. The screenshot provides evidence of the trunk state and the VLANs active on the switch.

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed sta

Switch(config)#do show interface trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/1     on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/1     1-1005

Port      Vlans allowed and active in management domain
Fa0/1     1,10,20

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     none

Switch(config)#
```

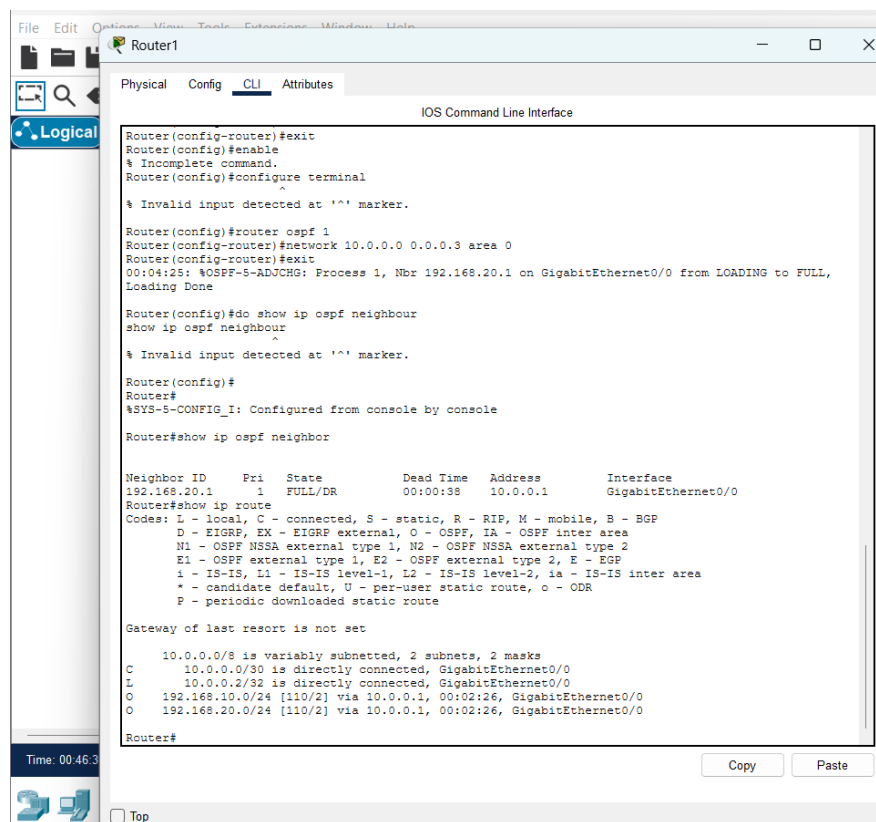
Figure 2: Switch Trunk Configuration Verification

6. OSPF Neighbor Verification

The completed topology contains two Cisco 2911 routers, a Layer 2 switch and four end devices. The routers are connected to each other through a routed link, while Router0 connects the local LAN to the switch. The end devices are divided between VLAN 10 and VLAN 20.

This topology was selected to demonstrate several practical networking concepts in one environment. The switch provides local Layer 2 connectivity, VLANs provide logical segmentation, Router0 performs inter-VLAN routing, and the router-to-router connection allows remote network reachability. Router1 represents another routed segment that can be reached using dynamic routing.

A topology diagram is also valuable documentation. In professional network work, a clear diagram helps administrators understand physical and logical relationships before troubleshooting. The screenshot shows the devices and links used for the practical implementation.



```
Router1
File Edit Options View Tools Extensions Window Help
Physical Config CLI Attributes
IOS Command Line Interface

Router(config-router)#exit
Router(config)#enable
% Incomplete command.
Router(config)#configure terminal
% Invalid input detected at '^' marker.

Router(config)#router ospf 1
Router(config-router)#network 10.0.0.0 0.0.0.3 area 0
Router(config-router)#exit
00:04:25: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.20.1 on GigabitEthernet0/0 from LOADING to FULL, Loading Done

Router(config)#do show ip ospf neighbour
show ip ospf neighbour
% Invalid input detected at '^' marker.

Router(config)#
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
192.168.20.1      1    FULL/DR         00:00:38    10.0.0.1       GigabitEthernet0/0

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       I - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0
L       10.0.0.2/32 is directly connected, GigabitEthernet0/0
O       192.168.10.0/24 [110/2] via 10.0.0.1, 00:02:26, GigabitEthernet0/0
O       192.168.20.0/24 [110/2] via 10.0.0.1, 00:02:26, GigabitEthernet0/0

Router#
```

Figure 3: OSPF Neighbor Verification on Router1

8. VLAN Configuration

OSPF, or Open Shortest Path First, is a link-state dynamic routing protocol. It allows routers to exchange information about reachable networks and calculate paths using the shortest-path-first algorithm. In this lab, OSPF process 1 was configured in area 0 on the routers.

Router0 advertised its local VLAN networks and the routed link. Router1 also participated in the same OSPF area. Once the OSPF adjacency reached the FULL state, the routers could exchange routing information. The routing table on Router1 shows OSPF-learned routes marked with the letter O.

Dynamic routing reduces the need to manually configure every remote route. It is particularly useful when a network contains multiple routers or when routes may change. The screenshot of the OSPF neighbor relationship and routing table is important evidence because it demonstrates that the routing protocol is not only configured but also operational.

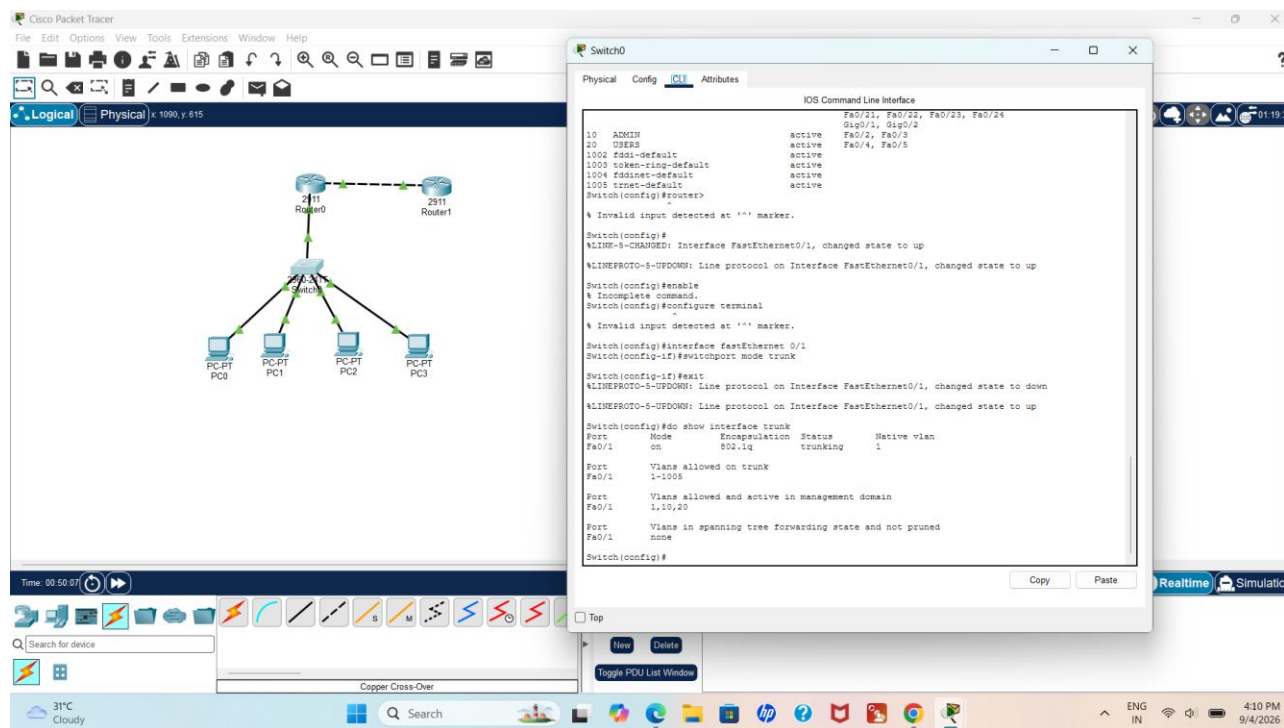


Figure 5: VLAN Configuration and VLAN Assignment on Switch

9. OSPF Routing Table

Access Control Lists (ACLs) are used on routers and Layer 3 devices to permit or deny traffic according to defined conditions. An extended ACL can evaluate parameters such as source IP address, destination IP address and protocol. In this lab, an extended ACL named USER-TO-ADMIN was created to control traffic from the user network toward the administration network.

The rule denies traffic from 192.168.20.0/24 to 192.168.10.0/24 and then permits other traffic. The ACL was applied inbound on the VLAN 20 router subinterface. Applying the policy at this point allows traffic from the user segment to be checked before it is routed onward.

ACLs are an important security control because they enforce network segmentation at Layer 3. Correct rule order is essential: ACL entries are evaluated from top to bottom, and an implicit deny exists at the end of an ACL. Therefore, policies should be planned carefully and verified after deployment.

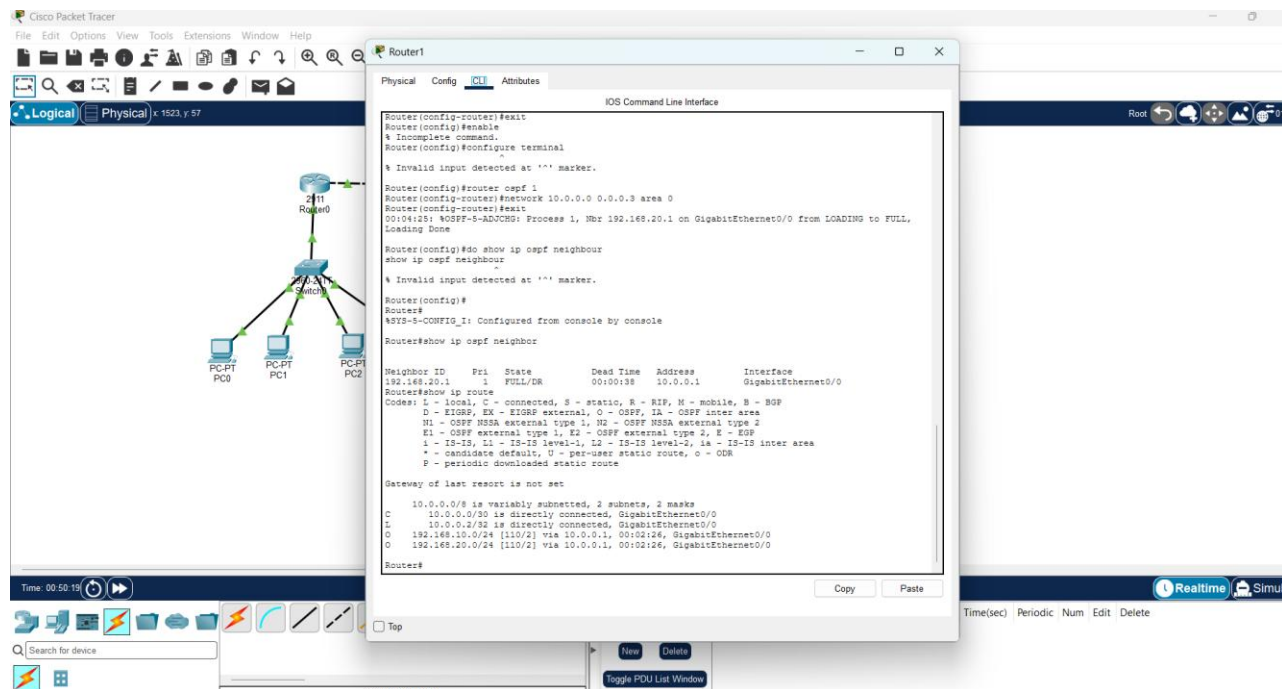


Figure 6: OSPF Configuration and Routing Table on Router0

10. Extended ACL Configuration

Connectivity testing is an essential part of network troubleshooting. A ping uses ICMP Echo Request and Echo Reply messages to test whether an IP destination can be reached. In this lab, a ping attempt to the administration network produced a Destination Host Unreachable response from the gateway.

This result is useful evidence because it demonstrates that communication between the tested networks was not successful at that point. The failure can be investigated by checking VLAN assignments, IP addressing, default gateways, routing information and ACL policies. In the context of this lab, the ACL was designed to restrict traffic from the user network toward the administration network, so the failed test can be used as a security-policy verification point.

A failed ping should not automatically be treated as a general network failure. ICMP may be blocked by policy, the destination may be unavailable, or a routing problem may exist. Troubleshooting therefore requires correlation with router and switch configuration.

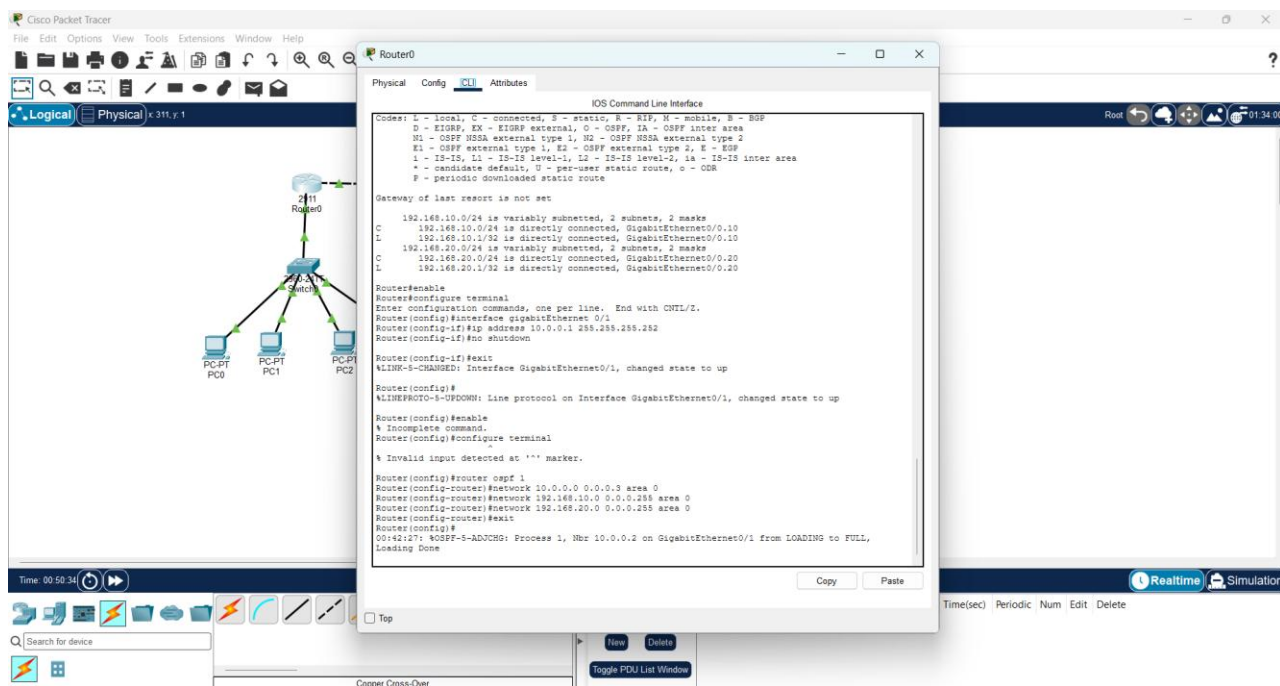


Figure 7: Extended ACL Configuration – USER-TO-ADMIN

11. Connectivity Testing

After configuring an ACL, it is important to verify the actual configuration and whether packets are matching the rules. The `show access-lists` command displays the ACL name, sequence numbers, configured conditions and match counters.

The evidence shows the USER-TO-ADMIN extended ACL and a deny statement targeting traffic from 192.168.20.0/24 toward 192.168.10.0/24. The match counter is useful because it indicates that traffic has reached the ACL entry. This is stronger evidence than simply showing the configuration command.

In a production network, ACL verification should be performed carefully because counters can help identify whether a security rule is actively being used. Administrators can compare expected traffic with observed matches and investigate unexpected behavior. This practical verification step connects configuration work with monitoring and troubleshooting.

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Figure 8: Connectivity Test – Ping and Destination Host Unreachable

12. ACL Verification

Wireshark is a packet analysis tool used to capture and inspect network traffic. In this lab, a live Wi-Fi capture was examined to identify different protocols and communication patterns. The screenshot shows QUIC traffic, including protected payload information. QUIC is a transport protocol commonly associated with modern encrypted application traffic and commonly operates over UDP.

Packet analysis is useful for understanding what is happening on a network at a protocol level. A capture can reveal source and destination addresses, ports, protocol types, packet lengths and timing information. Encrypted payloads may not expose application content, but metadata such as endpoints and protocol behavior can still be valuable for troubleshooting and monitoring.

The screenshot therefore serves as evidence that packet capture was performed and that protocol-level traffic was examined using Wireshark.

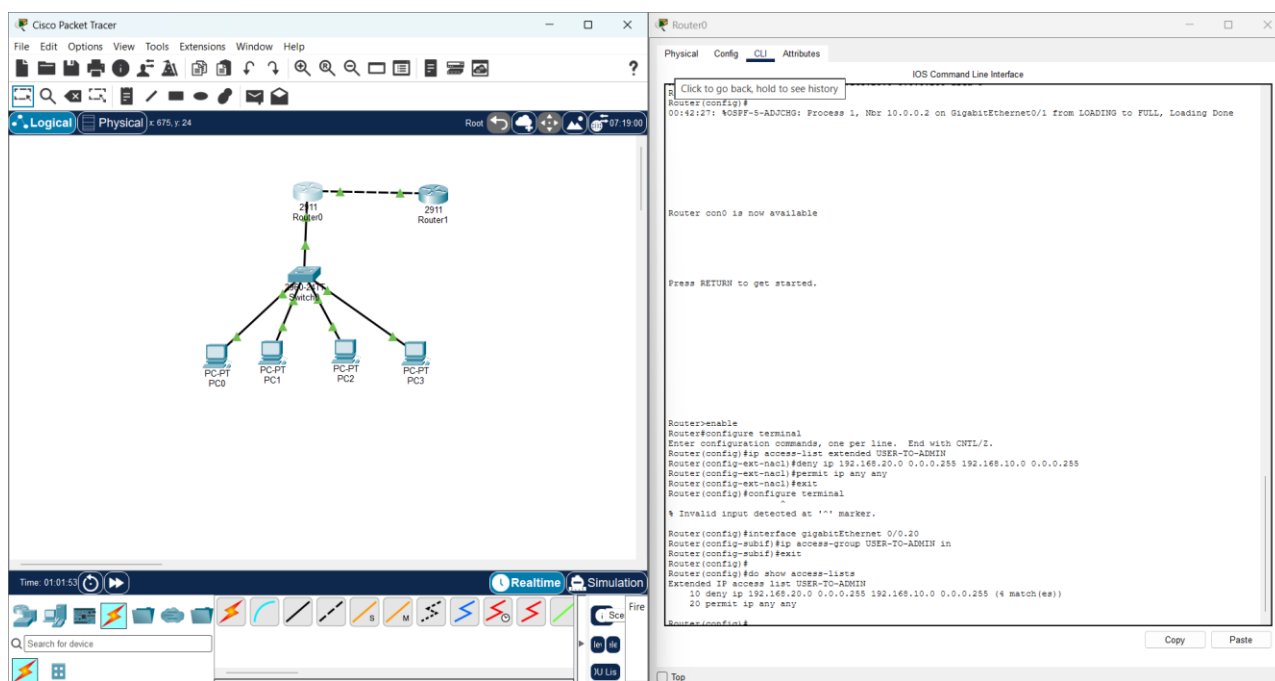


Figure 9: ACL Verification Using show access-lists

13. Wireshark QUICK Analysis

Display filters make large packet captures easier to investigate. The icmp filter was used to isolate Internet Control Message Protocol packets. The resulting view shows Echo Request packets from 10.180.197.187 to 8.8.8.8 and corresponding Echo Reply packets in the reverse direction.

This type of filtering is useful during connectivity troubleshooting because it removes unrelated TCP, DNS, QUIC and other traffic from the display. Analysts can then focus on the request/reply relationship, sequence numbers, packet timing and response behavior.

In a security operations or network operations workflow, display filters are frequently used to narrow an investigation. Examples include filtering by IP address, TCP/UDP port, DNS traffic, HTTP/TLS traffic or specific protocols. The screenshot demonstrates the practical use of an ICMP filter rather than only theoretical knowledge of packet analysis.

No.	Time	Source	Destination	Protocol	Length	Info
9253	97.465024900	2404:a800:6:133::e	2401:4900:bc73:fd8c...	QUIC	1292	Protected Payload (KP0)
9254	97.465212400	2404:a800:6:133::e	2401:4900:bc73:fd8c...	QUIC	1292	Protected Payload (KP0)
9255	97.465213700	2404:a800:6:133::e	2401:4900:bc73:fd8c...	QUIC	1292	Protected Payload (KP0)
9256	97.465484200	2401:4900:bc73:fd8c...	2404:a800:6:133::e	QUIC	98	Protected Payload (KP0), DCID=f3f2f399fa0262a1
9257	97.465997200	2404:a800:6:133::e	2401:4900:bc73:fd8c...	QUIC	1292	Protected Payload (KP0)
9258	97.465998700	2404:a800:6:133::e	2401:4900:bc73:fd8c...	QUIC	1292	Protected Payload (KP0)
9259	97.466100200	2404:a800:6:133::e	2401:4900:bc73:fd8c...	QUIC	1292	Protected Payload (KP0)
9260	97.467182600	2404:a800:6:133::e	2401:4900:bc73:fd8c...	QUIC	1292	Protected Payload (KP0)
9261	97.467184200	2404:a800:6:133::e	2401:4900:bc73:fd8c...	QUIC	1292	Protected Payload (KP0)
9262	97.467332800	2404:a800:6:133::e	2401:4900:bc73:fd8c...	QUIC	1292	Protected Payload (KP0)
9263	97.467333500	2404:a800:6:133::e	2401:4900:bc73:fd8c...	QUIC	1292	Protected Payload (KP0)
9264	97.467334800	2404:a800:6:133::e	2401:4900:bc73:fd8c...	QUIC	95	Protected Payload (KP0)
9265	97.467547800	2401:4900:bc73:fd8c...	2404:a800:6:133::e	QUIC	95	Protected Payload (KP0), DCID=f3f2f399fa0262a1
9266	97.488851900	2404:a800:6:133::e	2401:4900:bc73:fd8c...	QUIC	1292	Protected Payload (KP0)
9267	97.488866000	2404:a800:6:133::e	2401:4900:bc73:fd8c...	QUIC	677	Protected Payload (KP0)
9268	97.503660000	2401:4900:bc73:fd8c...	2404:a800:6:133::e	QUIC	96	Protected Payload (KP0), DCID=f3f2f399fa0262a1

```

> Frame 1: Packet, 1354 bytes on wire (10832 bits), 1354 bytes captured
> Ethernet II, Src: AzureWaveTec_0a:c9:11 (2c:3b:70:0a:c9:11), Dst:
> Internet Protocol Version 6, Src: 2401:4900:bc73:fd8c:554a:b920:52
> Transmission Control Protocol, Src Port: 62555, Dst Port: 443, Seq
0000 c2 73 4f 1b 3d 1d 2c 3b 70 0a c9 11 86 dd 60 04  :sO=,;,
0010 a2 10 05 14 06 3f 24 01 49 00 bc 73 fd 8c 55 4a  :...?$.
0020 b9 20 52 45 b6 0a 26 06 47 00 90 c2 7c bc c2 39  : RE..&
0030 09 b1 5f f2 75 c4 f4 5b 01 bb 31 85 b5 e0 2e 20  :...u-[
0040 2d ed 50 10 01 fd 3c b8 00 00 17 03 03 1f 4f 79  :-P...<
0050 1c da 65 fd 98 92 43 b2 2c 4d 12 54 2d 58 d1 da  :e...C
0060 ef 44 f7 dc 6a e5 7b 95 d1 e5 90 7e c5 0b bc 89  :-D-j-{
0070 1c b8 02 8a e7 d3 7b fd 41 a1 12 49 01 b7 04 a6  :...-{-
0080 f1 00 9d 4b 30 94 32 dd ed b0 e1 58 7c 2a ba 46  :...K0.2
0090 4e 29 9c ce 47 43 da 9e eb 8a 26 d9 a4 15 f7 b0  :N)...GC
00a0 a7 56 fb 7e 29 27 ba c2 ba e8 e1 10 b0 ed 03 01  :~V...)'.
00b0 21 9b 58 db 39 93 a3 c2 27 af dc b2 c8 c0 43 19  :!X.9...
00c0 7d 17 23 41 ba 25 9d de 7e d0 0a 39 ac 80 f0 5d  :}-#A.%-
00d0 01 d2 8e 04 71 76 7a 40 25 60 ac 1d 53 55 73 a1  :...qvz@
00e0 06 7b dd 01 b3 fd 75 ad 26 fa a2 e7 98 2a 01 17  :{-...u-
00f0 ed 11 b3 80 73 69 e8 0c fc dc 74 ce 17 e0 3f 57  :...s-
0100 92 e8 03 83 7a 71 15 0f 57 6d 06 2e 96 48 79 76  :...zq-
0110 ab 9c aa 8a 0e 33 6b 75 29 60 b1 33 d3 03 ac d0  :...3ku
0120 ce b1 6a ac e3 d3 e1 cb 11 c3 ef d8 4e cf 0b 15  :...j....
0130 16 0e 70 12 46 2e 55 4a f4 3f b6 93 b8 a1 1e 25  :-p.F.UJ
0140 0e 5d 07 96 e2 cf 1b 84 b0 c9 20 13 2b b9 0e 35  :l.....

```

Figure 10: Wireshark QUICK Traffic Capture and Packet Details

14. Wireshark ICMP Filtering

The selected ICMP packet is an Echo Request. The packet details show the IPv4 source and destination addresses and the ICMP fields including Type, Code, checksum, identifier and sequence number. These fields help an analyst understand the structure of an ICMP message.

An Echo Request is sent by the initiating host and an Echo Reply is returned by the destination when the request is handled successfully. Matching identifiers and sequence numbers help associate requests with replies. The capture also shows packet data carried within the ICMP message.

Examining individual packets is useful when a simple ping result is not enough. It allows an analyst to verify that requests are actually leaving the host, determine the destination, and check whether replies are returning. Wireshark provides this packet-level visibility without requiring the analyst to rely only on application output.

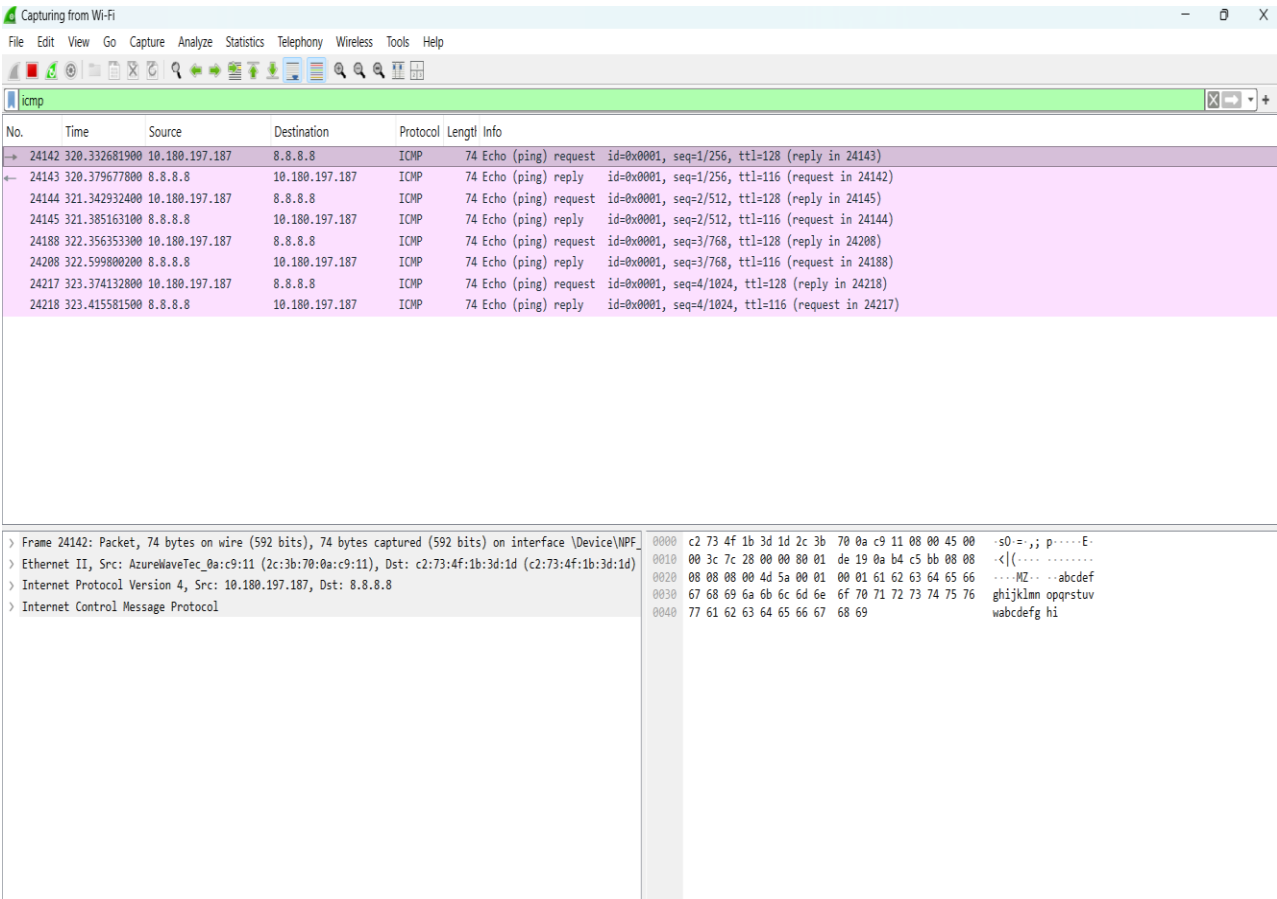


Figure 11: Wireshark ICMP Display Filter and Ping Traffic

15. ICMP Packet Inspection

TCP analysis provides visibility into connection-oriented network communication. The screenshot contains TCP packets involving port 443 and other ephemeral ports, together with ACK and application-data packets. TCP uses sequence numbers, acknowledgements and control flags to provide reliable delivery.

Port 443 is commonly associated with HTTPS/TLS traffic. When encrypted application protocols are used, the actual application payload may not be readable, but TCP metadata remains visible. Analysts can still inspect endpoints, ports, sequence and acknowledgement numbers, window information, retransmissions and connection behavior.

This kind of analysis is useful in troubleshooting slow connections, identifying unusual communication patterns and understanding how applications communicate across a network. The screenshot demonstrates that the capture contained multiple TCP exchanges and that the packets could be inspected at the protocol level.

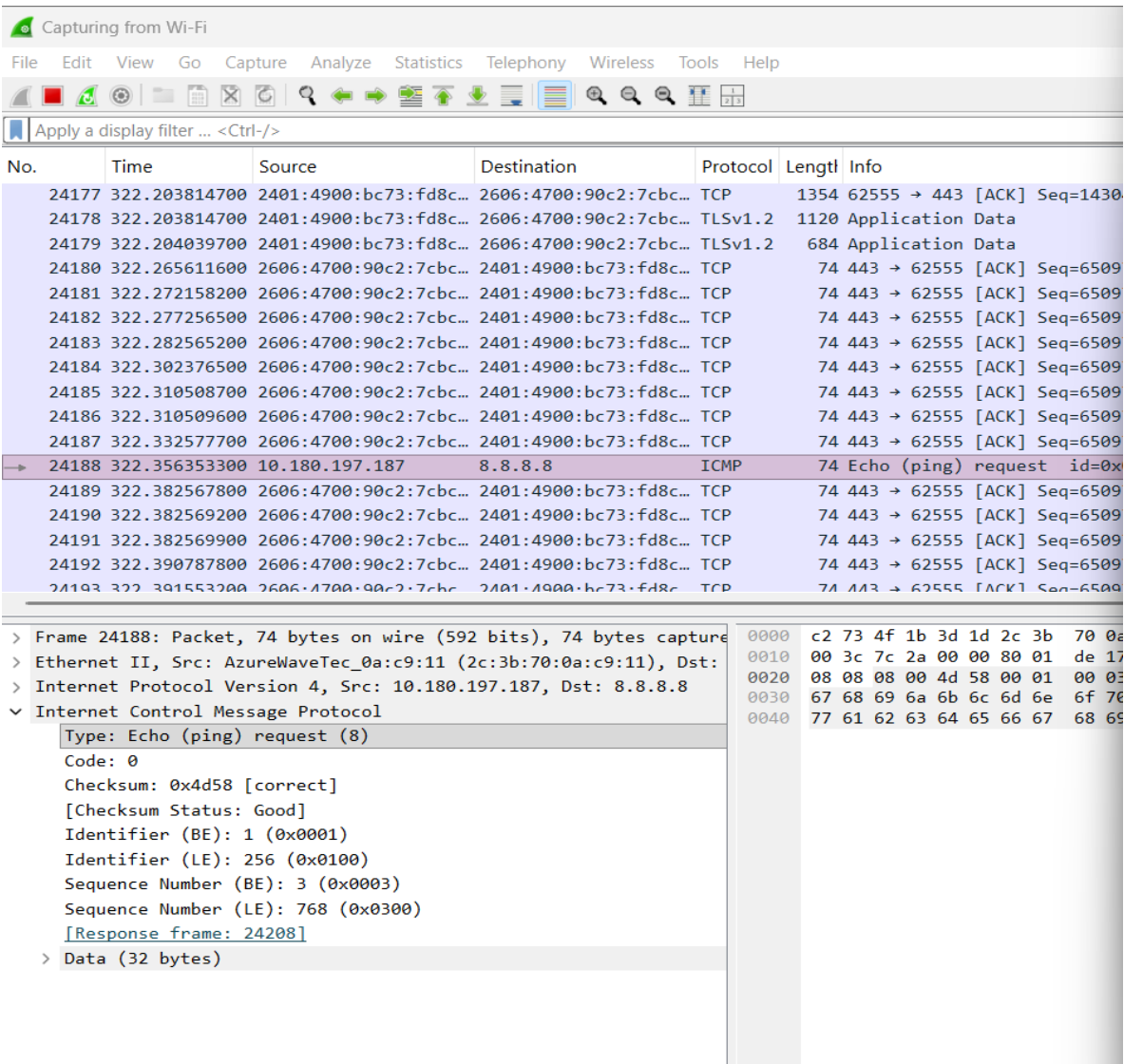


Figure 12: Wireshark ICMP Echo Request/Reply Packet Analysis

16. TCP Traffic Analysis

DNS converts domain names into IP addresses so that applications can locate network services. The Wireshark capture shows DNS queries and responses between the local host and a DNS server. The entries include queries related to chatgpt.com and other domains.

DNS analysis is useful in both networking and cybersecurity. From a network perspective, DNS helps identify whether name resolution is working correctly. From a security perspective, unusual DNS activity can sometimes indicate suspicious software behavior, command-and-control communication or attempts to reach unexpected domains.

The screenshot demonstrates the complete query/response concept: a client sends a request for a domain record and the DNS server responds with the relevant record information. Wireshark allows these exchanges to be inspected and filtered, making DNS an important protocol for practical network monitoring.

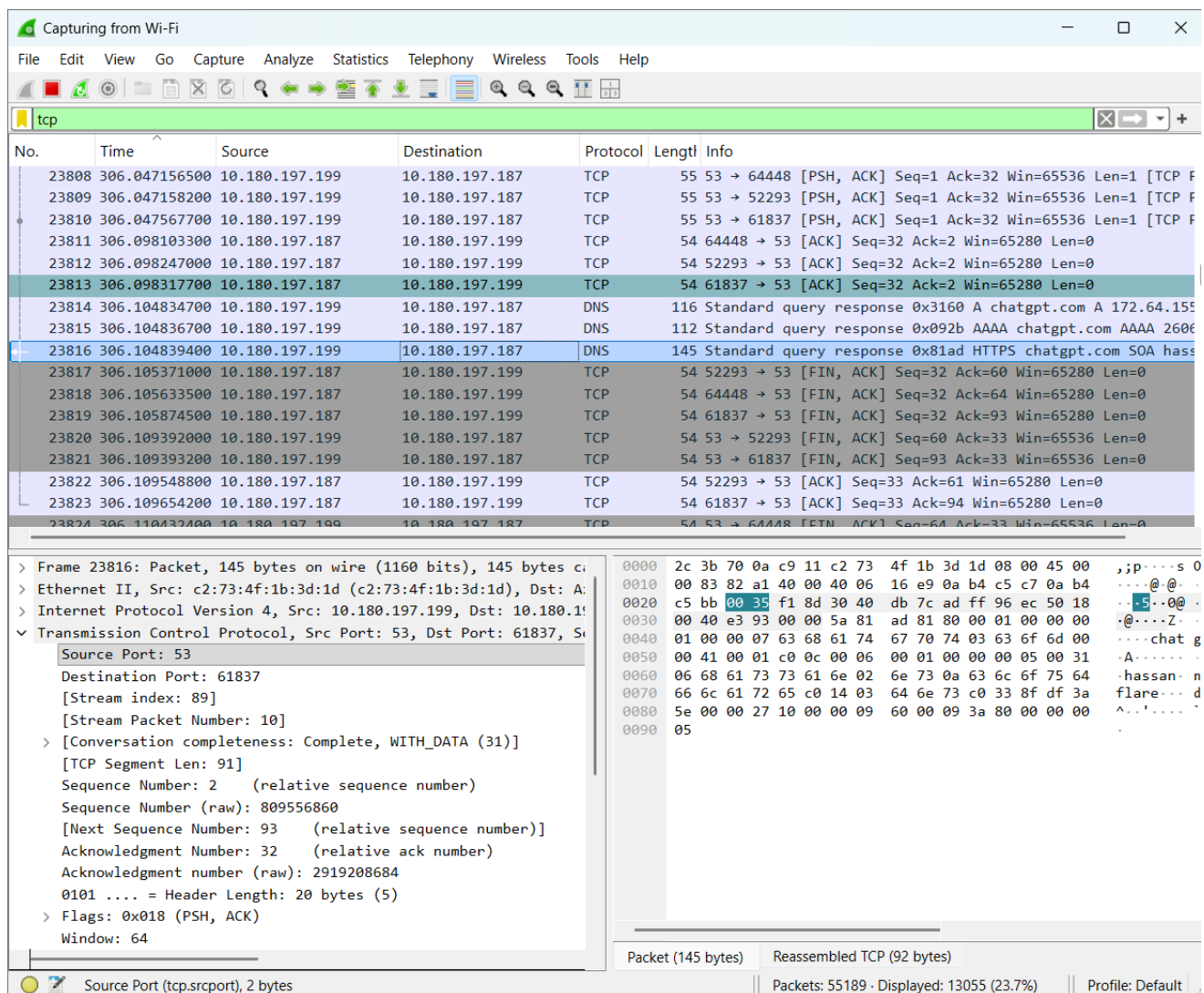


Figure 13: Wireshark TCP Traffic Capture

17. DNS Traffic Analysis

A TCP packet can be examined in detail to understand how the transport layer manages communication. The screenshot shows fields including source port, destination port, stream index, sequence number, acknowledgement number, TCP flags and window information.

The source port is an ephemeral client-side port while the destination port identifies the service being contacted. Sequence and acknowledgement numbers allow TCP endpoints to track transmitted and received data. Flags such as ACK and PSH provide information about how the segment is being handled.

Detailed TCP inspection is particularly useful during troubleshooting. An analyst can investigate retransmissions, resets, handshake problems, abnormal connection termination and unusual port activity. Combining packet details with higher-level protocol information provides a more complete view of network behavior.

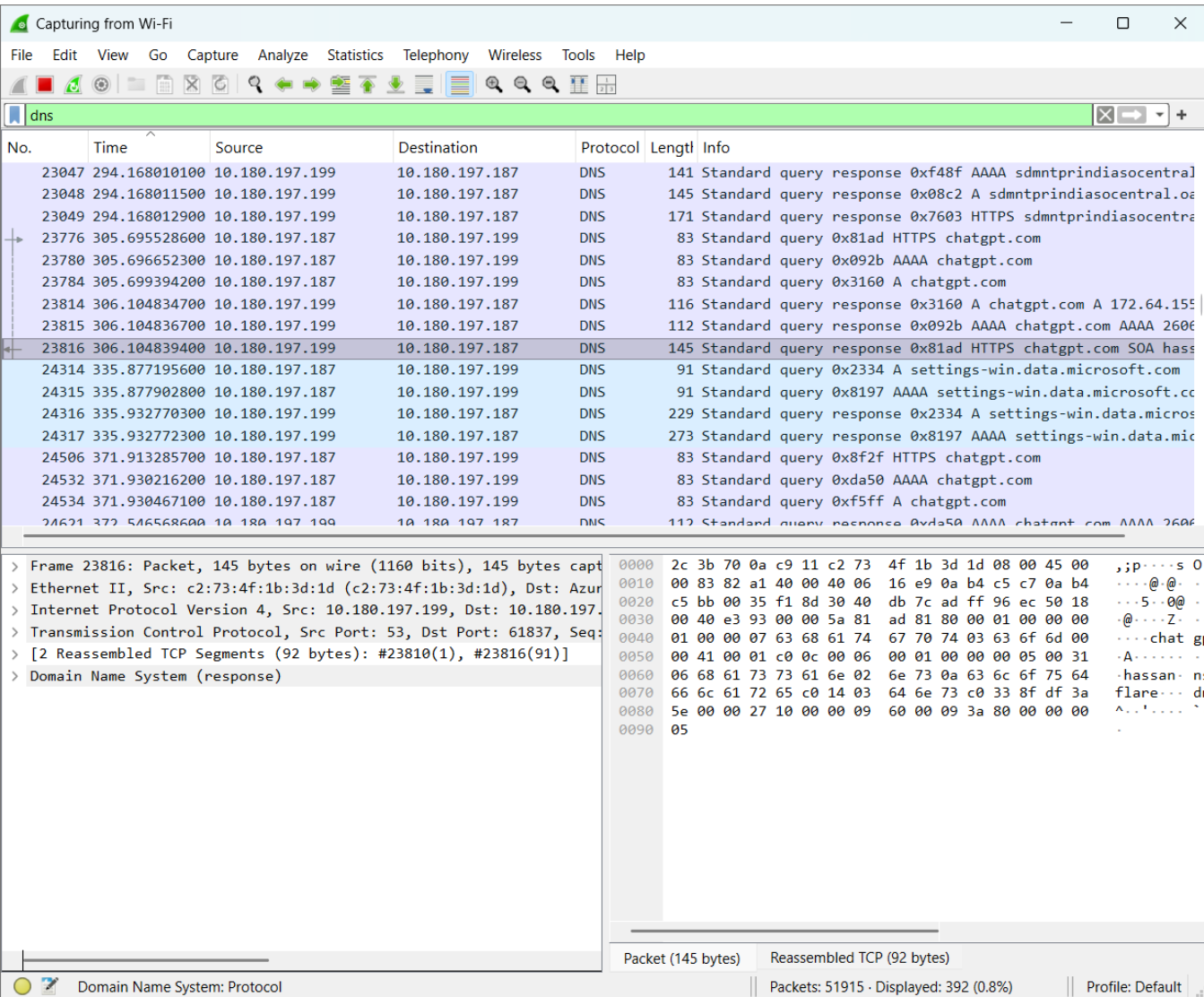


Figure 14: Wireshark DNS Query and Response Analysis

18. TCP Packet Detail Analysis

Kali Linux was used to configure and verify the UFW firewall. UFW, or Uncomplicated Firewall, provides a simplified interface for managing Linux firewall rules. The screenshot shows the firewall enabled and active, with logging enabled and the default incoming policy set to deny while outgoing traffic is allowed.

An SSH rule for TCP port 22 is explicitly allowed. This demonstrates a basic firewall-hardening approach: block unsolicited inbound traffic by default and permit only services that are required. Such a policy reduces the exposed attack surface of a host.

Firewall status verification is an important operational step. The command output provides direct evidence of the active policy and configured rules. In a real environment, firewall rules should be reviewed regularly and limited to trusted sources where appropriate. The lab demonstrates the fundamental relationship between host hardening, access control and network security.

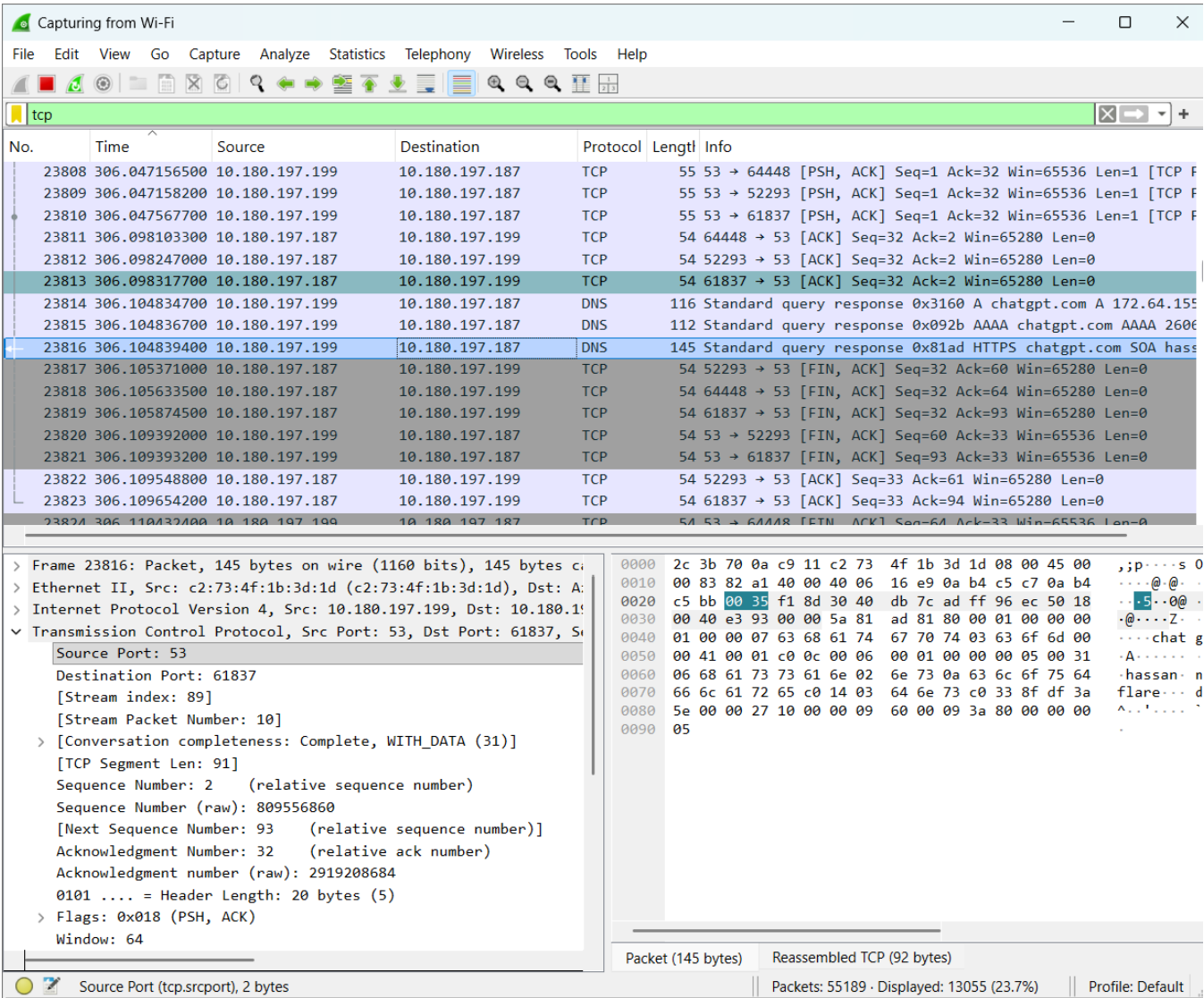
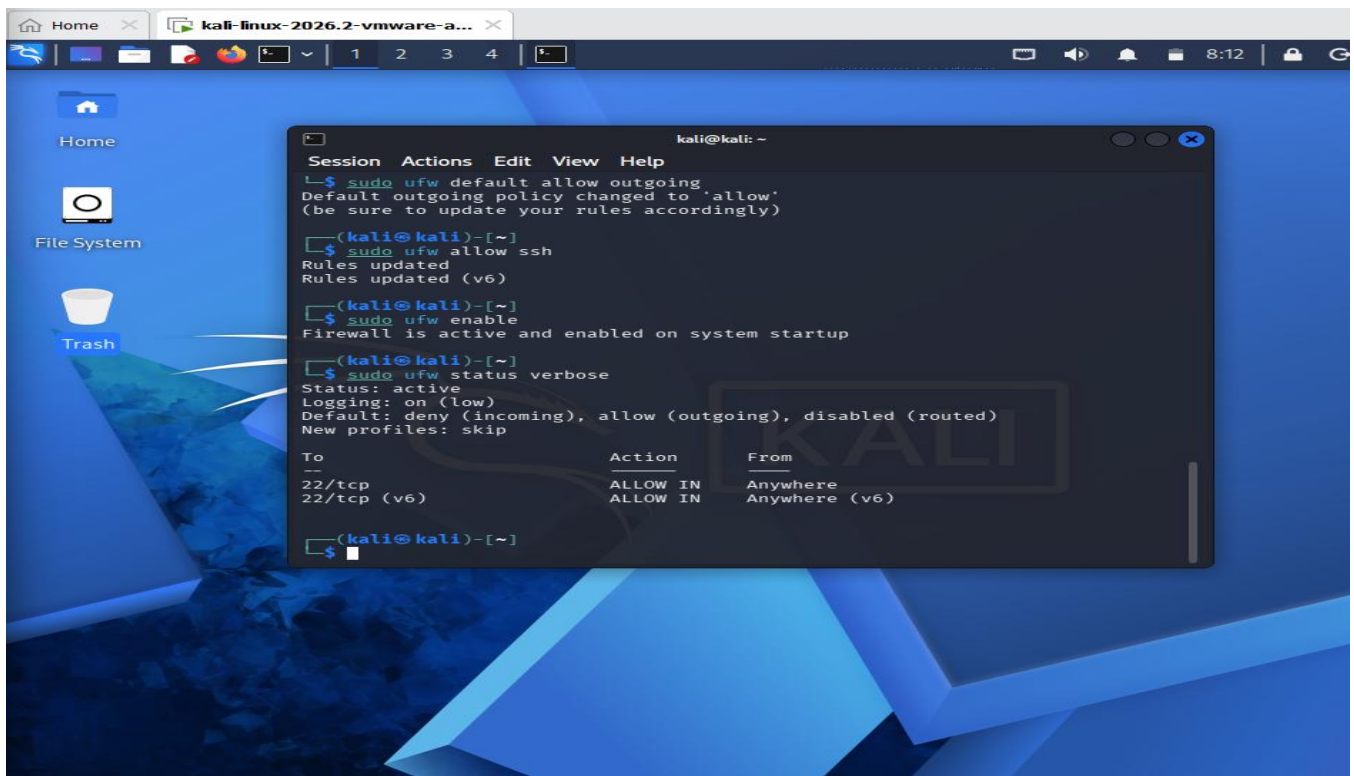
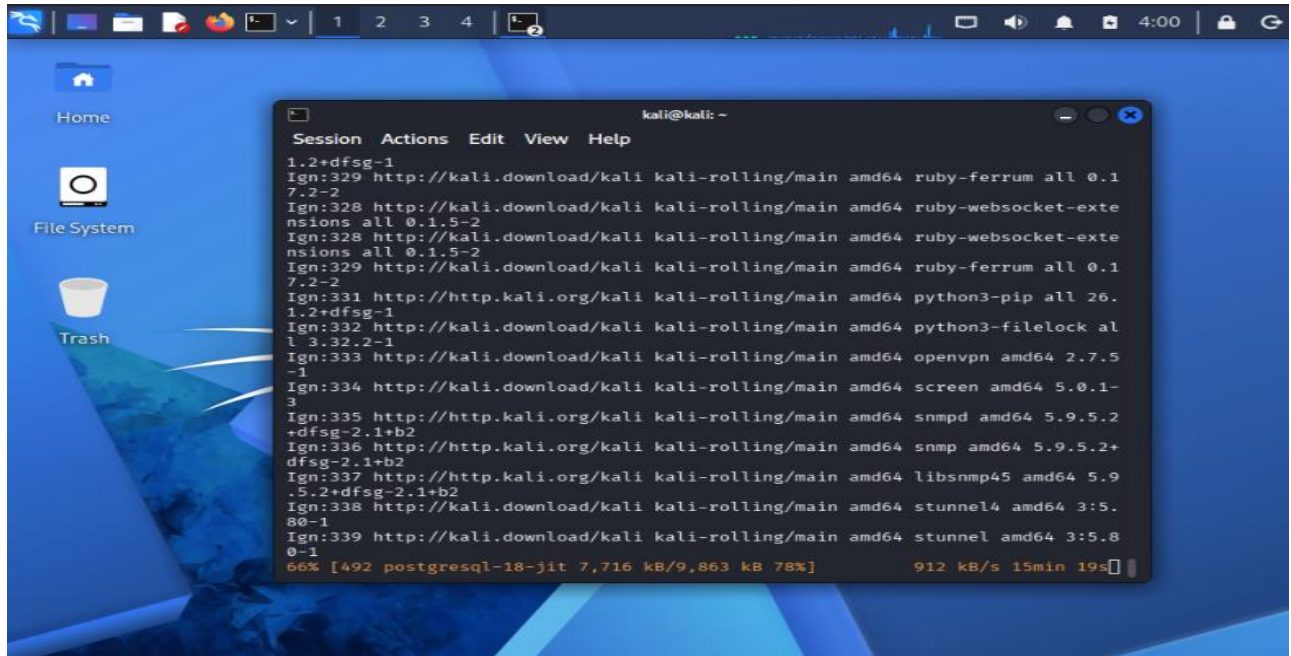


Figure 15: Wireshark TCP Packet Detail and Port Analysis

Screenshot of kali linux:



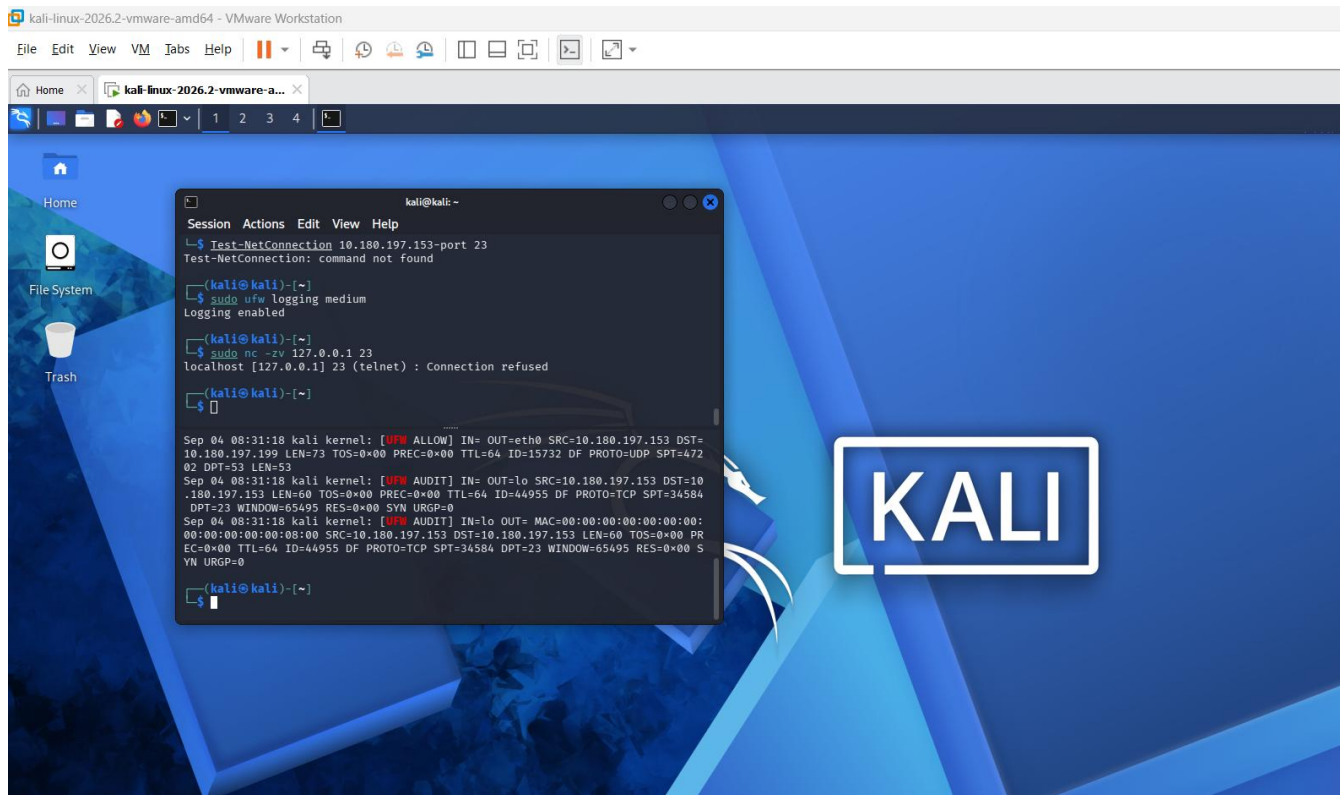
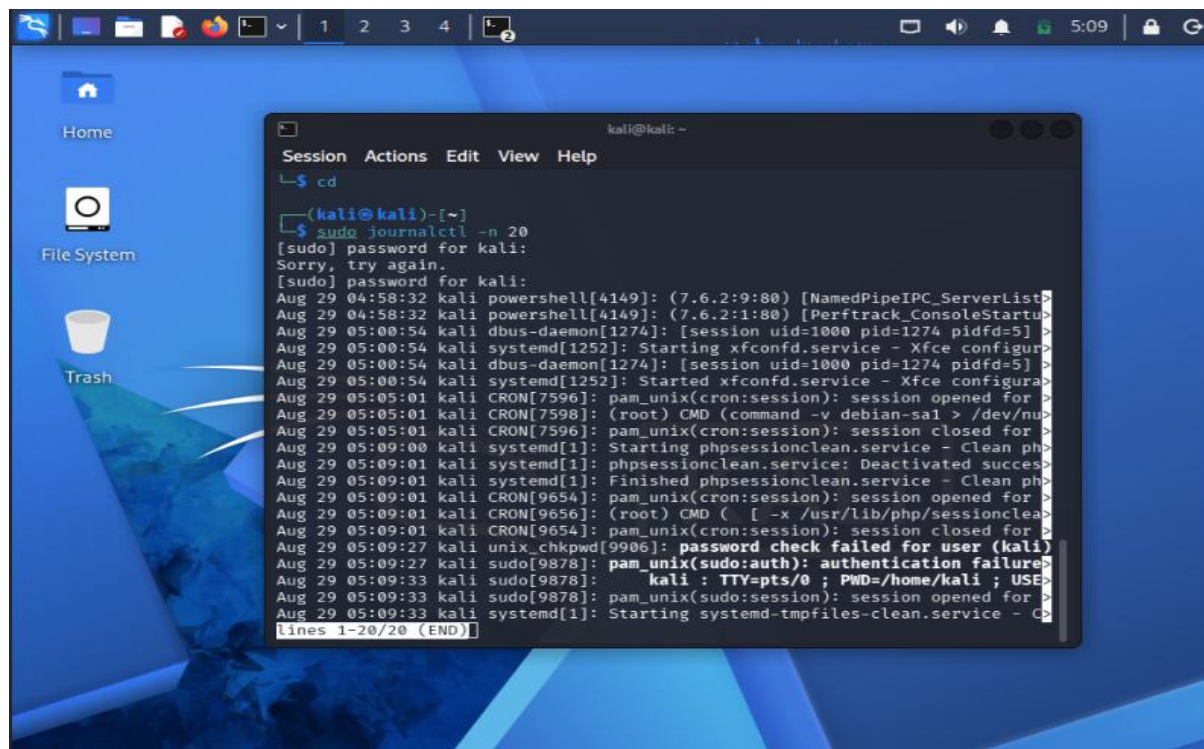


Figure16 : Kali Linux UFW installation & Firewall Status



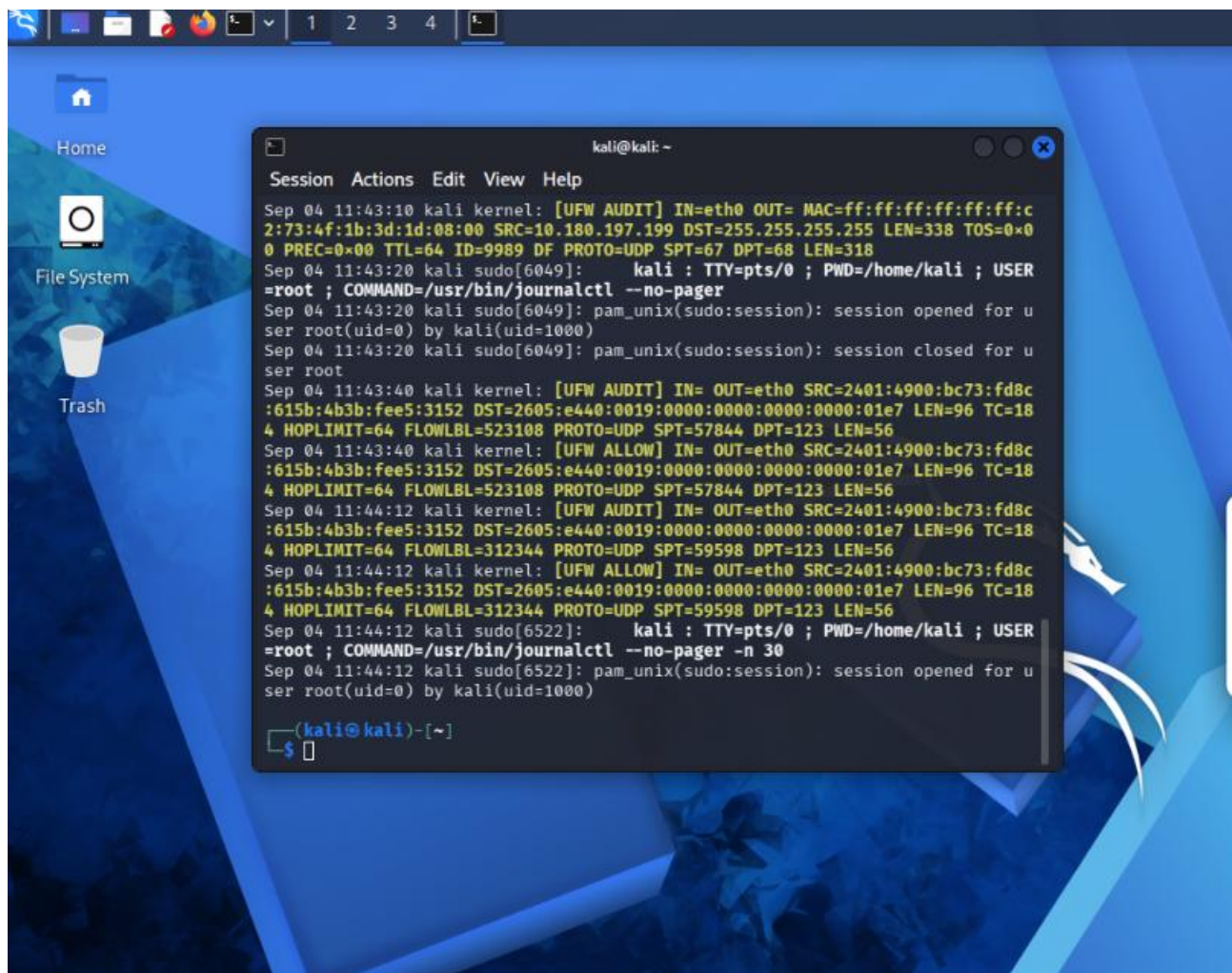
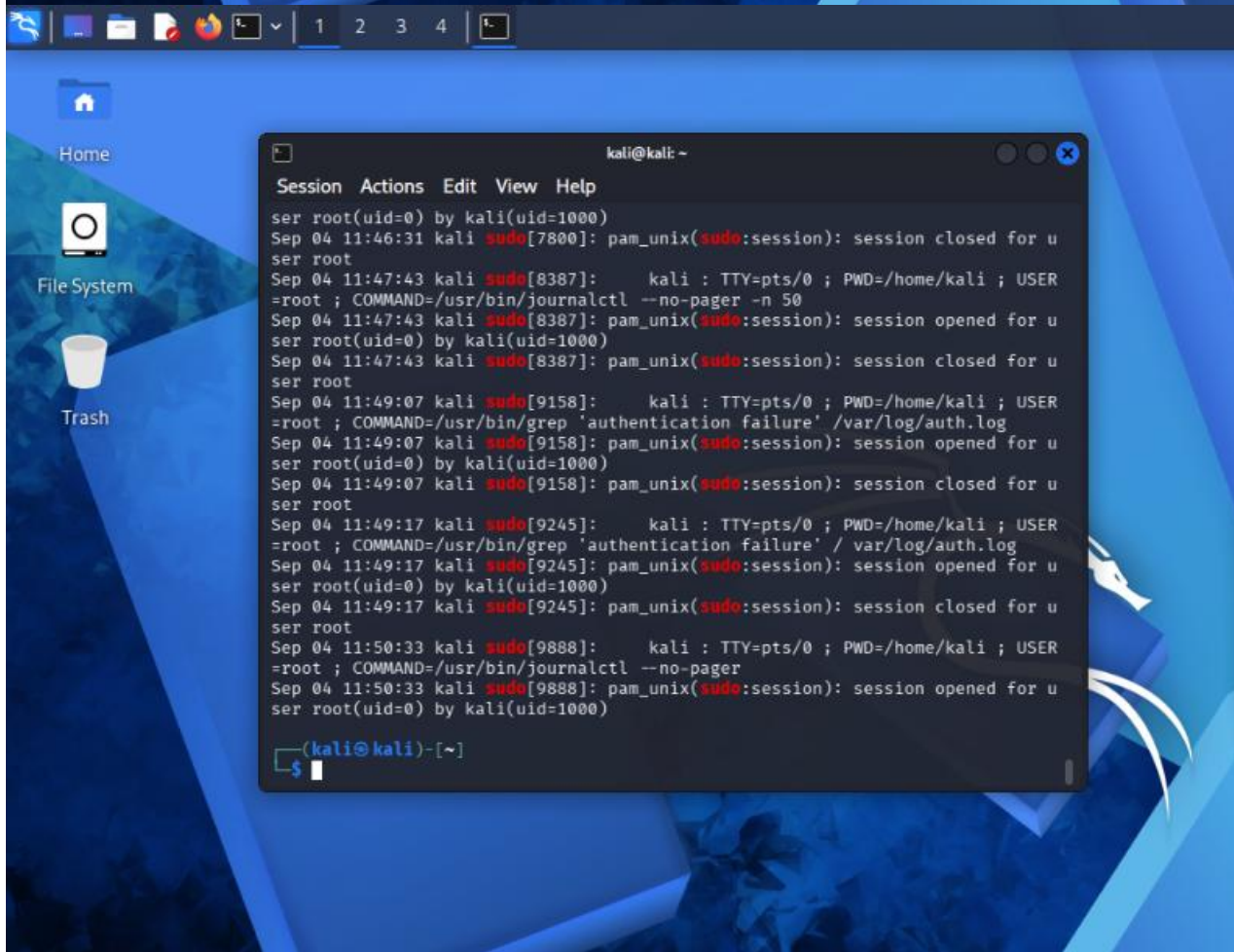
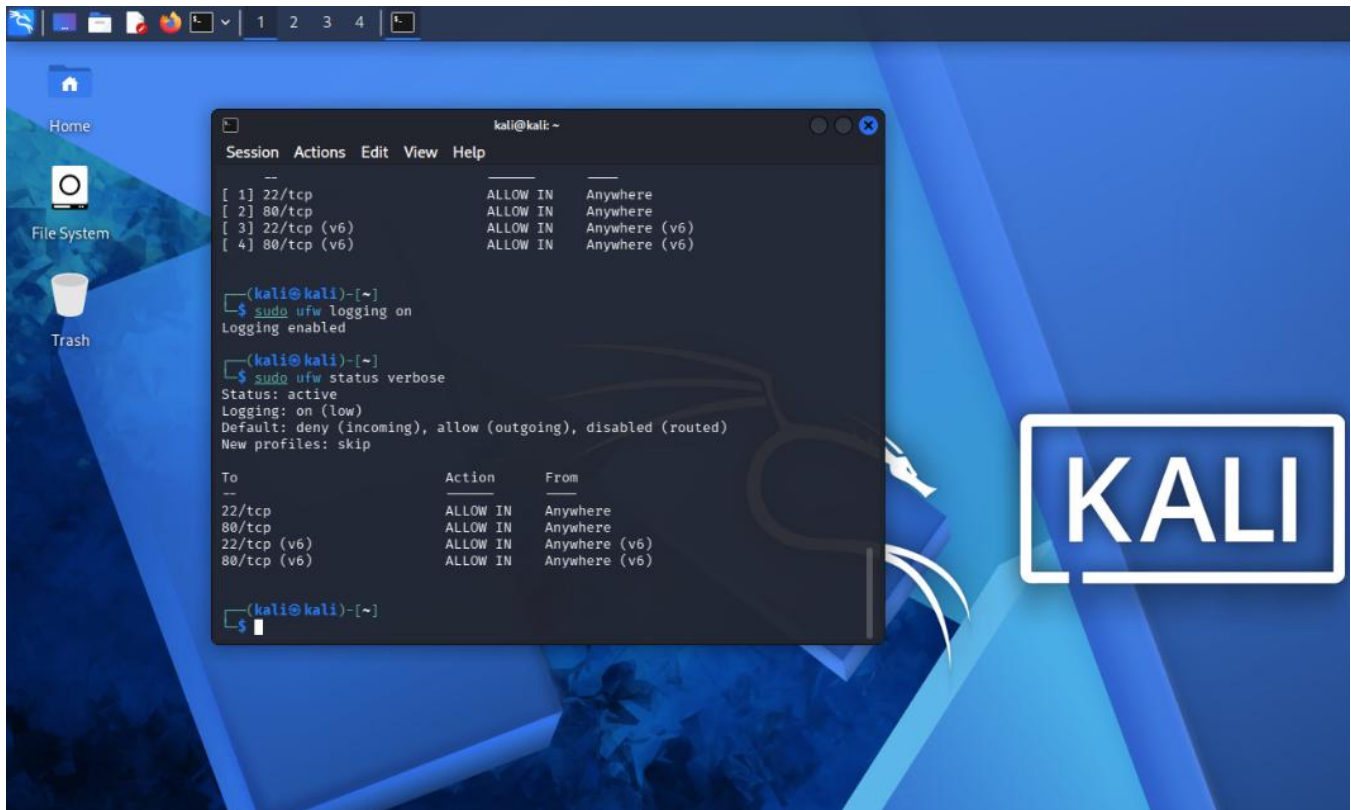


Figure 18: System Journal Log Analysis & UFW Firewall Audit Logs



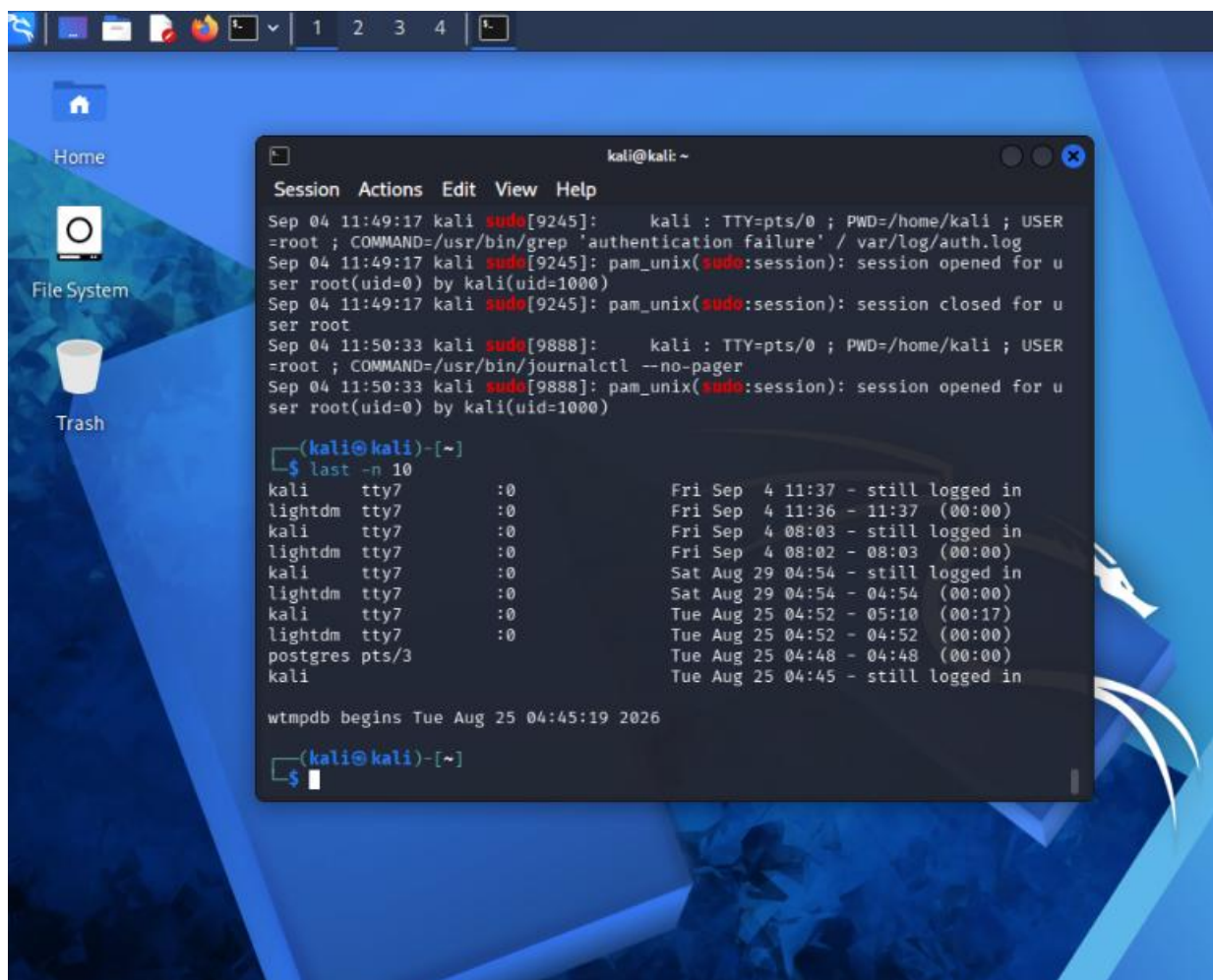


Figure 19: Authentication and Sudo Session Logs & Security Event and Firewall Log Monitoring

20. Conclusion

This project provided hands-on exposure to practical network administration, routing, "segmentation and security controls. The network was built using Cisco Packet Tracer and "included VLAN segmentation, inter-VLAN routing, OSPF dynamic routing and an extended ACL.

"Verification commands and connectivity tests were used to validate the configuration and "to demonstrate how a security rule can affect traffic between networks.

Wireshark analysis added a packet-level monitoring component to the project. ICMP, TCP, DNS and QUIC traffic were filtered and inspected to understand communication at different protocol layers. This is directly relevant to network troubleshooting and SOC/NOC monitoring because

"packet captures can provide evidence when logs or application-level errors are not sufficient.

Kali Linux and UFW were also used to demonstrate host-level firewall hardening. The final result is "

"a combined networking and cybersecurity lab that demonstrates configuration, verification, "

"monitoring and troubleshooting skills rather than only theoretical knowledge. The collected "

screenshots provide practical evidence of the work completed.